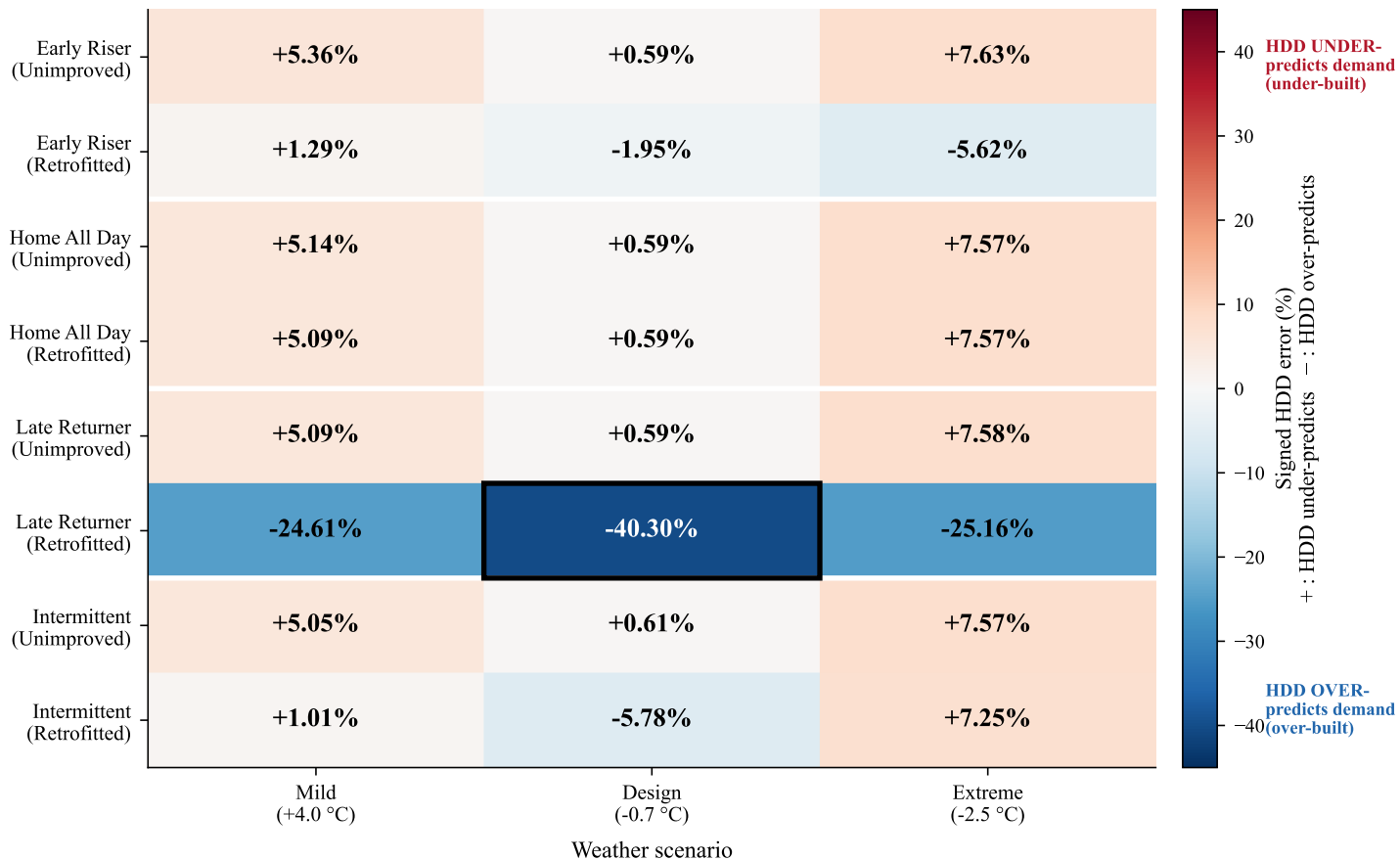


HDD prediction error across the behaviour × fabric × weather grid

Worst over-prediction: HDD says 5.81 kW · sim says 4.14 kW → HDD over-builds by 40%



At design conditions with unimproved fabric, peak demand is heat-loss-limited; archetype effects collapse to within rounding.